



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 3 Name: Dhanish Ladwani PRN: 24070521020

1. Experiment Title:

Develop a web application using HTML and Javascript to validate a Password using conditional statements and loops.

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

3. Experiment Program Code:

Case Study 3.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Registration Portal</title>
  <link rel="stylesheet" href="Case Study 3.css">
</head>

<body>
  <div class="container">
    <div class="card">
      <h2>Registration</h2>
      <form id="registerForm" onsubmit="handleRegister(event)">
        <div class="form-group">
          <label for="registerUser">Username</label>
          <input type="text" id="registerUser" placeholder="Choose a username" required>
        </div>
        <div class="form-group">
          <label for="registerEmail">Email</label>
          <input type="text" id="registerEmail" placeholder="Enter email address" required>
        </div>
        <div class="form-group">
          <label for="registerPassword">Password</label>
          <input type="password" id="registerPassword" placeholder="Create strong password"
required>
        </div>
        <div class="form-group">
          <label for="registerConfirmPassword">Confirm Password</label>
          <input type="password" id="registerConfirmPassword" placeholder="Confirm your
password" required>
        </div>
        <div class="checkbox-group">
          <input type="checkbox" id="showRegPassword" onclick="toggleRegVisibility()">
          <label for="showRegPassword">Show Passwords</label>
        </div>
        <input type="submit" class="btn" value="Register">
      </form>

      <div id="message"></div>
    </div>
  </div>
```

```

        <footer style="text-align: center; margin-top: 30px; font-size: 14px; color: gray; border-top: 1px
solid lightgray; padding-top: 15px;">
            Made by Dhanish Ladwani PRN - 24070521020
        </footer>
    </div>

    <script src="Case Study 3.js"></script>
</body>

</html>

```

Case Study 3.css

```

body {
    background-color: whitesmoke;
    color: black;
    font-family: sans-serif;
    margin: 0;
    padding: 0;
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
}

.container {
    width: 100%;
    max-width: 400px;
    padding: 20px;
}

.card {
    background-color: white;
    padding: 30px;
    border: 1px solid lightgray;
    border-radius: 12px;
    box-shadow: 0 4px 12px lightgray;
}

h2 {
    color: darkblue;
    text-align: center;
    margin-top: 0;
    margin-bottom: 20px;
}

form {
    display: flex;
    flex-direction: column;
    gap: 15px;
}

.form-group {
    display: flex;
    flex-direction: column;
    gap: 5px;
}

.checkbox-group {
    display: flex;
    align-items: center;
    gap: 8px;
}

.checkbox-group label {
    font-weight: normal;
    font-size: 14px;
    color: black;
}

```

```

    cursor: pointer;
}

.checkbox-group input {
    cursor: pointer;
}

label {
    font-weight: bold;
    font-size: 14px;
    color: dimgray;
}

input[type="text"],
input[type="password"] {
    padding: 10px;
    border: 1px solid lightgray;
    border-radius: 6px;
    font-size: 16px;
}

input[type="text"]:focus,
input[type="password"]:focus {
    border-color: blue;
    outline: none;
}

.btn {
    padding: 12px;
    background-color: royalblue;
    color: white;
    border: none;
    border-radius: 6px;
    font-size: 16px;
    font-weight: bold;
    cursor: pointer;
    transition: background-color 0.2s;
}

.btn:hover {
    background-color: blue;
}

#message {
    margin-top: 20px;
    padding: 10px;
    border-radius: 6px;
    font-weight: bold;
    text-align: center;
    display: none;
}

#message.error {
    background-color: mistyrose;
    color: red;
    border: 1px solid tomato;
}

#message.success {
    background-color: honeydew;
    color: green;
    border: 1px solid limegreen;
}

```

Case Study 3.js

```

function getPasswordValidationError(password) {
    if (password.length == 0) {

```

```

        return "Please enter a password.";
    }
    if (password.length < 8) {
        return "Password is too short (minimum 8 characters).";
    }
    if (password.length > 20) {
        return "Password is too long (maximum 20 characters).";
    }
    if (password.indexOf(" ") !== -1) {
        return "Password must not contain spaces.";
    }

    var hasUppercase = false;
    var hasLowercase = false;
    var hasNumber = false;
    var hasSpecial = false;
    var specialChars = "!@#$%^&*(),.?\\\":{}|<>";

    for (var i = 0; i < password.length; i++) {
        var char = password[i];
        if (char >= 'A' && char <= 'Z') {
            hasUppercase = true;
        } else if (char >= 'a' && char <= 'z') {
            hasLowercase = true;
        } else if (char >= '0' && char <= '9') {
            hasNumber = true;
        } else if (specialChars.indexOf(char) !== -1) {
            hasSpecial = true;
        }
    }

    if (!hasUppercase) {
        return "Password must contain at least one uppercase letter.";
    }
    if (!hasLowercase) {
        return "Password must contain at least one lowercase letter.";
    }
    if (!hasNumber) {
        return "Password must contain at least one number.";
    }
    if (!hasSpecial) {
        return "Password must contain at least one special character.";
    }

    return null;
}

function handleRegister(event) {
    event.preventDefault();
    var user = document.getElementById("registerUser").value.trim();
    var email = document.getElementById("registerEmail").value.trim();
    var password = document.getElementById("registerPassword").value;
    var confirmPassword = document.getElementById("registerConfirmPassword").value;
    var message = document.getElementById("message");

    if (user === "") {
        message.innerText = "Please choose a username.";
        message.className = "error";
        message.style.display = "block";
        return;
    }
    if (email === "") {
        message.innerText = "Please enter your email.";
        message.className = "error";
        message.style.display = "block";
        return;
    }
    if (email.indexOf("@") === -1) {
        message.innerText = "Please enter a valid email address.";
        message.className = "error";
    }

```

```

        message.style.display = "block";
        return;
    }

    var passwordError = getPasswordValidationError(password);
    if (passwordError) {
        message.innerText = passwordError;
        message.className = "error";
        message.style.display = "block";
        return;
    }

    if (password !== confirmPassword) {
        message.innerText = "Passwords do not match.";
        message.className = "error";
        message.style.display = "block";
        return;
    }

    message.innerText = "Registration successful!";
    message.className = "success";
    message.style.display = "block";
}

function toggleRegVisibility() {
    var passwordField = document.getElementById("registerPassword");
    var confirmPassword = document.getElementById("registerConfirmPassword");

    if (passwordField.type === "password") {
        passwordField.type = "text";
        confirmPassword.type = "text";
    } else {
        passwordField.type = "password";
        confirmPassword.type = "password";
    }
}

```

4. Output:

Registration Portal

File C:/Users/Admin/Downloads/Javascript/Experiment%203/case%20study%203/Case%20Study%203.html

Ask Gemini

Registration

Username
Dhanish

Email
dhanishladwani@gmail.com

Password
Dhanish@0711

Confirm Password
Dhanish@0711

☒ Show Passwords

Register

Registration successful!

Made by Dhanish Ladwani PRN - 24070521020

Registration Portal

File C:/Users/Admin/Downloads/Javascript/Experiment%203/case%20study%203/Case%20Study%203.html

Ask Gemini

Registration

Username
Dhanish

Email
dhanishladwani@gmail.com

Password
Dhanish0711

Confirm Password
Dhanish0711

☒ Show Passwords

Register

Password must contain at least one special character.

Made by Dhanish Ladwani PRN - 24070521020

5. Result/Conclusion:

A registration web page was developed to validate a password using conditional statements and a for loop. The loop scans each character of the password to check for the presence of uppercase letters, lowercase letters, numbers and special characters, while conditional statements enforce length limits (8-20 characters) and disallow spaces. Appropriate error messages are displayed for each validation failure, and a success message is shown once the password and confirm-password fields match all criteria. This case study reinforced the use of loops, string traversal and conditional logic for form validation in JavaScript.